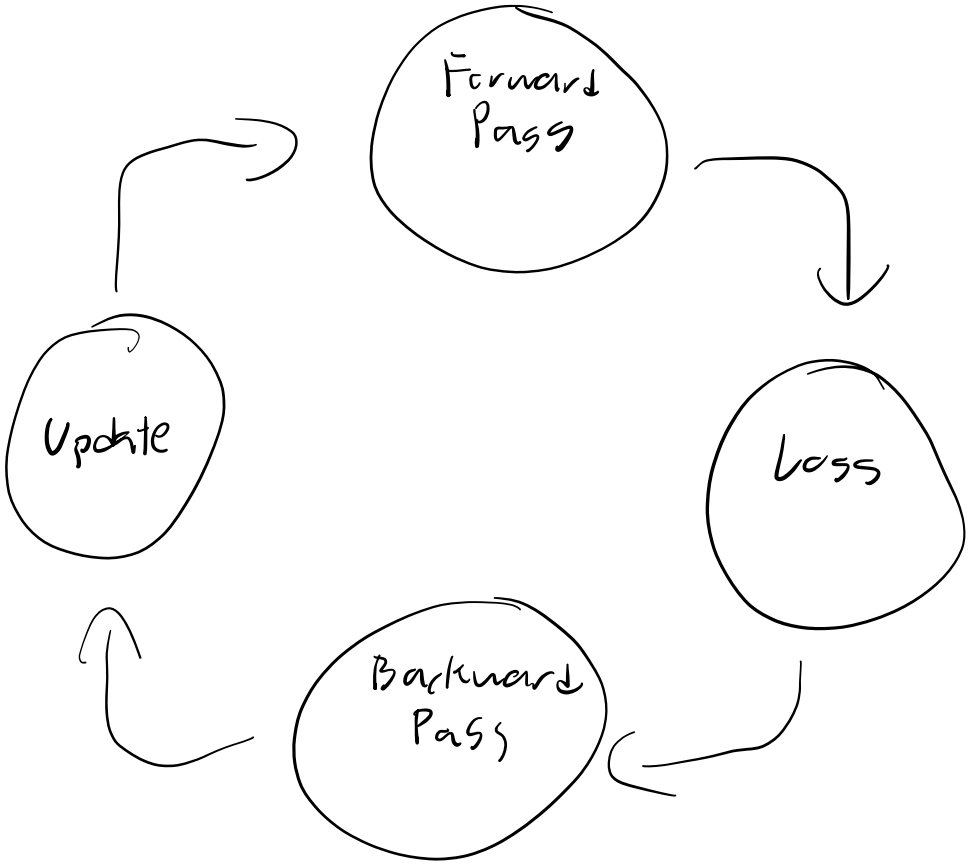
Goal: Find weights so network's output matches the correct label for each point

Original Data, (x_1, x_2) and labels y stay constant

Weights and biases get adjusted

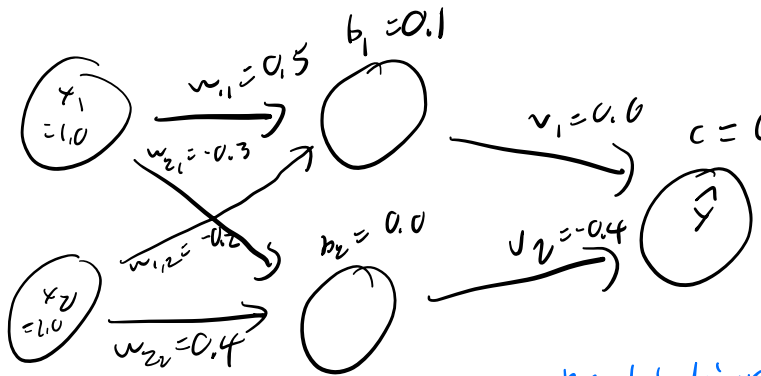


2 inputs → 2 hidden neurons → 1 output

$x_1 = 1.0$

$x_2 = 2.0$

$y = +1$



1. Forward Pass and Compute prediction

First Neuron

$$a_1 = w_{11}x_1 + w_{12}x_2 + b_1 = (0.5)(1) + (-0.2)(2) + 0.1 = 0.2$$

$$h_1 = \tanh(0.2) = 0.1974$$

Second Neuron $a_2 = w_{21} x_1 + w_{22} x_2 + b_2 = (-0.3)(1) + (0.4)(2)$

$= 0.5$

$h_2 = \tanh(0.5) = 0.4621$

Output Neuron $s = v_1 h_1 + v_2 h_2 + c = 0.0336$

$\hat{y} = \tanh(0.0336) = 0.0336$

but

$y = +1$

2. Get Loss

MSE $L = (\hat{y} - y)^2 = (0.0336 - 1)^2 = 0.9340$

3. Backward Pass

$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) = 2(0.0336 - 1) = -1.9329$

$\frac{d}{ds} \tanh(s) = 1 - \tanh^2(s)$

$\frac{\partial L}{\partial s} = \frac{\partial L}{\partial \hat{y}} \cdot (1 - \hat{y}^2) = -1.9307$

$\frac{\partial L}{\partial v_1} = \frac{\partial L}{\partial s} \cdot h_1 = -0.3811$

$\frac{\partial L}{\partial v_2} = -0.8922$

$\frac{\partial L}{\partial c} = \frac{\partial L}{\partial s} \cdot 1 = -1.9307$

$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial s} \cdot v_1$ $\frac{\partial L}{\partial h_2} = \frac{\partial L}{\partial s} \cdot v_2$

$$\frac{\partial L}{\partial a_1} = \frac{\partial L}{\partial h_1} \cdot (1 - h_1^2) \qquad \frac{\partial L}{\partial a_2} = \frac{\partial L}{\partial h_2} \cdot (1 - h_2^2)$$

then same for $\frac{\partial L}{\partial w_{11}}$, $\frac{\partial L}{\partial w_{12}}$, $\frac{\partial L}{\partial b_1}$

$\frac{\partial L}{\partial w_{21}}$, $\frac{\partial L}{\partial w_{22}}$, $\frac{\partial L}{\partial b_2}$

4. Update

$$p_{new} = p_{old} - \alpha \cdot \frac{\partial L}{\partial p}$$

Ex. $w_{12} = -0.2 - (0.5)(-2.2206)$

So weights get updated

5. Repeat 1-4 w/ new weights